

NGO TRAN DINH DUY

Ho Chi Minh City University of Technology – VNU-HCM (HCMUT)
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PROFESSIONAL SUMMARY

- Third-year student majoring in Computer Science at Ho Chi Minh City University of Technology – VNU-HCM, focusing on AI/ML, Quantum Computing and applied research.
- Undergraduate Student at **HPCC Team** (High Performance Computing Center).

EDUCATION

Ho Chi Minh City University of Technology (HCMUT - VNU-HCM) 2023 – 2027
(Expected) Computer Science (English-Taught Program) | Major: Artificial Intelligence.

HPCC Team (High Performance Computing Center) 2025 – Present
Undergraduate Student

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, SQL, Bash.
- **Machine Learning / Deep Learning:** scikit-learn, TensorFlow, Keras, PyTorch, YOLO (Ultralytics), Bi-LSTM, Transformer, NumPy, Pandas, Matplotlib.
- **Signal & Data Processing:** MNE (EEG/MEG).
- **MLOps & Development Tools:** Docker, Linux, Git, GitHub, Jupyter Notebook, Weights & Biases (WandB).
- **Databases & Networking Tools:** MySQL, Cisco Packet Tracer.
- **Languages:** English (IELTS Academic 6.5).

EXPERIENCE

AI Engineer Intern — Viettel High Tech Jun 2026 – Aug 2026
AI Ops for Autonomous Radio Access Networks

- Contributed to the design and development of an AI Ops framework for autonomous RAN operations, covering fault management, network optimization through traffic steering, and RAN energy saving.
- Processed network telemetry, alarms, performance counters, topology, traffic, mobility, and QoE/KPI data to support anomaly detection, root cause analysis, impact assessment, and operational decision-making.
- Prototyped AI/ML-based analytics for detecting network anomalies, identifying performance degradation, and recommending corrective actions under network KPI, QoS, and capacity constraints.
- Developed closed-loop automation workflows for RAN parameter adjustment, traffic steering, fault recovery, and energy-saving actions, including post-action monitoring and performance verification.
- Explored energy optimization strategies such as cell or carrier shutdown, sleep/wake scheduling, and transmission-power reduction while maintaining service quality requirements.

Technologies: Python, AI/ML, AI Ops, Radio Access Networks, Anomaly Detection, Root Cause Analysis, Time-Series Analytics, Traffic Steering, Network Optimization.

PROJECTS

Agentic AI Assistant for University Regulations | [GitHub](#) Jan. 2026 – Present

- Engineered a sophisticated Retrieval-Augmented Generation (RAG) chatbot using Google Gemini (1.5/2.0 Pro/Flash) and Milvus/Zilliz vector database to automate student queries regarding complex

university regulations.

- Developed an automated ETL and monitoring pipeline using Selenium and hybrid PDF extraction (PyMuPDF + Gemini Vision) to ensure high-fidelity document indexing and real-time Discord notifications for regulatory updates.
- Optimized system reasoning by implementing metadata-enriched retrieval and pioneered research into migrating from traditional vector databases to a Knowledge Graph (KG) architecture to better model hierarchical regulatory data.

Multimodal ML Benchmarking and Analytics Platform | [GitHub](#) *Jan. 2026 – May 2026*

- Built JSON-driven Chart.js/Plotly dashboards for tabular, SQuAD, SVHN, and Flickr8k EDA, surfacing validated scale and quality signals across 238K+ samples.
- Implemented leakage-aware ad-click ML with sklearn Pipeline/ColumnTransformer, CV target encoding, RandomizedSearchCV, isotonic calibration, threshold tuning, bootstrap CI, and slice audits; verified 0.9885 test ROC-AUC and 97.5% tuned-threshold accuracy.
- Benchmarked TF-IDF/BoW/char-hash classifiers, DistilBERT, ResNet-18, and MobileNetV2; verified 99.99% SQuAD question-type accuracy and 96.32% SVHN accuracy.

Geospatial Object Detection from Satellite Imagery | [Kaggle](#) *Nov. 2025 – Jan. 2026*

- Participated in a computer vision competition aimed at identifying and localizing rural buildings from high-resolution satellite images.
- Fine-tuned YOLO object detection models; implemented data augmentation and preprocessing pipelines to handle complex geographical terrains; optimized bounding box predictions to significantly improve the Mean Average Precision (mAP) score.

Skeleton-Based Action Recognition for Elderly Care | [GitHub](#) *Feb. 2026 – Present*

- Developed a real-time fall detection system utilizing deep learning and computer vision.
- Extracted 3D human skeleton landmarks using MediaPipe to ensure privacy and computational efficiency on edge devices; designed and trained an LSTM architecture to process temporal coordinate sequences, accurately classifying critical fall events.
- Achieved 78.75% validation accuracy on a binary fall-detection task with an LSTM architecture (64 hidden units, single layer, 0.5 dropout), demonstrating robust real-time inference capabilities in resource-constrained environments.

Tutor Support System Project | [Frontend](#) | [Backend](#) | [Demo](#) *Nov. 2025 – Dec. 2025*

- Built a Spring Boot 3 (Java 17) backend for a university tutoring platform, with REST APIs for tutor discovery, profile viewing, and session slot booking.
- Implemented core academic support modules for enrollments, support needs, subjects, and staff management using layered architecture (controller/service/repository), DTO mapping, and validation.
- Prepared the system for real deployment and integration with PostgreSQL/JPA, OpenAPI documentation, Docker-based local setup, and test scripts for endpoint verification.